

ELLINGTON FIELD, TX, USA

WMO: 722436

Lat: 29.617N

Lon: 95.167W

Elev: 32

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.7	36.6	18.6	14.0	41.9	23.4	17.7	44.7	24.2	59.6	21.8	56.6	10.3	340	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	97.1	78.0	94.9	78.4	93.0	78.4	82.4	88.7	81.4	87.5	80.8	87.2	8.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.3	162.7	84.2	80.6	158.9	83.8	79.2	151.8	83.1	46.2	89.7	44.9	88.2	44.2	87.5	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	18.3	16.7	29.0	100.8	1.7	3.9	27.8	103.6	26.8	105.8	25.9	108.0	24.6	110.8		
			26.3	85.2	2.0	2.5	24.8	87.0	23.7	88.4	22.6	89.8	21.2	91.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.7	55.4	58.0	62.7	70.0	77.4	82.4	84.5	84.6	80.7	72.7	63.4	55.9
	DBStd	12.70	9.61	9.11	7.48	6.27	4.66	2.81	2.23	2.94	4.68	7.02	8.21	9.02
	HDD50	138	55	26	8	0	0	0	0	0	0	0	7	42
	HDD65	1159	320	228	131	28	1	0	0	0	0	20	124	307
	CDD50	7700	222	252	400	600	849	973	1070	1073	921	705	411	224
	CDD65	3247	23	33	59	178	385	523	605	608	471	260	77	25
	CDH74	32081	73	120	250	1040	3375	5728	7162	7090	4765	1972	426	80
	CDH80	13378	2	11	27	201	1207	2512	3400	3392	1991	580	52	3
Wind	WSAvg	7.9	9.3	9.3	8.9	9.4	8.6	7.8	6.6	6.0	6.2	7.1	7.8	8.4
Precipitation	PrecAvg	53.2	4.3	3.0	3.7	4.1	5.2	5.8	4.5	4.3	4.7	5.4	4.2	4.1
	PrecMax	79.5	10.5	9.1	8.5	9.8	13.2	24.6	12.1	9.1	11.5	17.6	12.7	8.3
	PrecMin	23.7	0.6	0.5	0.8	0.1	0.0	0.3	1.4	0.5	1.6	0.0	0.7	1.1
	PrecStd	13.0	2.4	2.2	2.1	3.0	3.3	4.8	2.9	2.6	2.3	4.5	2.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	81.6	82.4	87.9	93.5	96.8	99.5	99.5	99.7	91.2	84.4	80.7
		MCWB	68.7	68.6	68.4	72.7	77.0	78.1	76.9	77.9	76.7	77.5	75.5	70.4
	2%	DB	75.8	77.5	80.5	84.3	90.5	93.4	96.6	97.1	94.7	88.4	81.4	76.9
		MCWB	68.1	68.2	68.8	71.8	76.1	78.1	77.9	78.5	77.8	76.0	71.5	69.6
	5%	DB	72.9	74.4	77.1	82.0	88.5	91.4	94.3	94.9	91.3	86.0	79.0	73.2
		MCWB	66.9	66.0	67.7	70.9	76.0	78.1	78.4	78.8	76.4	75.5	70.4	68.0
	10%	DB	70.3	71.6	73.6	80.5	86.4	90.4	92.6	92.9	90.0	82.5	76.6	71.3
		MCWB	66.0	65.6	66.7	70.3	75.5	78.0	78.6	79.0	76.4	74.0	69.1	67.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.5	71.6	74.0	77.1	80.7	82.3	83.3	83.8	83.6	81.6	77.5	73.3
		MCDB	75.6	77.1	79.4	83.8	86.5	88.6	89.4	90.4	89.8	85.0	81.6	76.7
	2%	WB	70.5	70.2	72.3	75.3	79.1	81.2	81.9	82.4	81.4	79.6	75.4	71.7
		MCDB	73.5	74.5	76.7	80.7	85.7	87.5	88.3	89.4	86.1	83.7	78.9	75.1
	5%	WB	69.2	69.0	70.8	74.1	78.1	80.3	81.0	81.4	80.3	78.3	73.5	70.0
		MCDB	72.0	72.4	74.5	78.8	84.8	86.7	87.9	88.0	85.2	82.6	77.1	73.0
	10%	WB	67.0	67.4	69.5	73.0	77.2	79.4	80.2	80.8	79.2	76.9	71.6	67.6
		MCDB	69.7	70.4	72.8	77.4	83.9	86.0	87.5	87.6	84.4	81.2	75.3	70.5
Mean Daily Temperature Range	5% DB	MDBR	17.6	16.9	17.8	17.0	16.9	16.0	16.9	17.1	17.0	17.1	17.8	17.7
		MCDBR	20.0	18.7	19.4	17.8	18.2	17.6	19.1	19.2	18.9	17.8	19.3	19.4
		MCWBR	13.8	10.9	10.9	8.5	7.3	6.1	6.4	6.0	6.6	8.1	12.0	13.2
	5% WB	MCDBR	17.3	15.1	15.7	14.2	15.9	16.0	17.0	16.6	15.2	14.3	16.3	17.3
		MCWBR	13.6	10.2	11.1	7.8	7.2	6.6	6.7	6.2	6.7	7.9	12.9	14.0
Clear-Sky Solar Irradiance	taub		0.343	0.362	0.376	0.439	0.481	0.473	0.499	0.486	0.458	0.382	0.372	0.353
	taud		2.508	2.467	2.417	2.236	2.157	2.232	2.179	2.218	2.263	2.468	2.441	2.488
	Ebn at Noon		283	287	289	272	260	261	254	257	260	276	269	273
	Edh at Noon		29	32	36	45	49	45	47	45	42	32	30	28
All-Sky Solar Radiation	RadAvg		915	1033	1417	1691	1891	1960	1893	1815	1576	1381	1044	813
	RadStd		88	176	120	122	133	154	178	150	166	167	106	96

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+169

Nomenclature: See separate page